

What Happens in Paris, Does Not Stay in Paris: Trade Fairs and Search and Matching Frictions

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Abstract

Search and matching frictions prevent firms from forming international trade linkages. Firms commonly use trade fairs to overcome these frictions. However, despite often being subsidized by export promotion agencies, we lack evidence on how trade fairs facilitate link formation. We exploit a unique feature of Hungarian firms’ participation in the 1900 Paris World Exhibition, where a trial exhibition revealed firms’ ex-ante export potential category to develop a novel bounding strategy that compares treated firms to control groups from “above” and “below” in export potential. To implement our empirical strategy, we collected and digitized a novel dataset detailing firms’ exhibition attendance and export status, linking entities across five different datasets. We find that participation increases export probability by 8.5–14% and employment by 15.4–23% over eight years. Effects are larger when firms face fewer domestic competitors and more potential international buyers. This highlights both matching benefits and congestion effects when search and matching frictions are reduced. To carry out this estimation, we build a novel historical panel dataset of Hungarian firms for the 1894 to 1906 period by digitizing and combining over 12,000 firm records from five historical datasets.

Keywords: buyer-supplier links, export promotion, trade fairs, search frictions, industrial policy, economic history

JEL Codes: F14, N73, O14, L25

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1 Introduction

Exporting and importing require firms to search for and connect with foreign buyers and suppliers. These firm-to-firm links are often hindered by substantial search and matching frictions. For example, a McKinsey 2021 report suggests that identifying a single suitable supplier can be costly and take several months (McKinsey & Company, 2021). Understanding the nature and magnitude of these frictions, as well as how they can be alleviated, is central to our understanding of international trade, the organization of global production networks, and the propagation of shocks across borders.

We contribute to this literature by studying one of the most common channel firms use to facilitate link formation: trade fairs. Indeed, export promotion agencies in at least 103 countries provide support to exporting firms (Lederman et al., 2010), with trade fair participation among the most common interventions. Unlike interventions that provide individual firms with targeted access to new buyers or suppliers (e.g., Atkin et al. 2017; Cai et al. 2024; Buus et al. 2025; Munch and Schaur 2018; Carballo and Volpe Martincus 2008), trade fairs bring together large numbers of firms simultaneously, exposing participants not only to potential partners but also to competitors, potentially generating congestion effects. At the same time, in contrast to transport and communication infrastructure improvements that broadly reduce effective distance between all buyers and sellers (e.g., Jensen 2007; Jensen and Miller 2018; Allen 2014; Bernard et al. 2019), trade fairs limit participation to a selected subset of firms, allowing for a more focused environment for link formation.

We overcome the selection bias that complicates the evaluation of trade fair participation by exploiting a unique historical experiment: the Hungarian government’s selection process for the 1900 Paris World Exhibition. Our setting offers three advantages. First, the government used a preceding national exhibition as a “trial run,” where an independent jury awarded firms based on their export potential. Second, constraints such as limited pavilion space, the desire for balanced industrial and regional representation, and transport costs of exhibits prevented the government from sending only the firms with the highest export potential. Third, the limited communication infrastructure of the era—costly travel and reliance on postal mail—meant firms had few alternatives to in-person meetings for establishing international contacts, allowing us to isolate the exhibition’s impact from other correlated, private search channels that abound in modern settings.

The 1900 Paris *Exposition Universelle*, which attracted 48 million visitors and featured 70,000 businesses from 56 countries, was one of the largest international commercial gatherings in history. It served as a showcase for recent innovations, including the diesel engine, moving sidewalk, motion picture, and passenger trolleybus. In 1900, Hungary was part of the Austrian Hungarian Monarchy (AHM), a polity with central foreign policy and customs union, but devolved finances, economic and industrial policies.

Recognizing world fairs as an opportunity to promote Hungarian industry, the government began preparations 13 years in advance. A central element was the *1896 Budapest Millennial Exhi-*

bition, organized as a “trial run” to identify firms with the greatest export potential. The 1896 fair provided broad coverage of Hungarian firms, and an independent jury assessed each exhibit’s export potential, awarding four prize categories. Based on this, a separate selection committee—distinct from the jury—was tasked with choosing firms for Paris. While the primary goal was to promote exports by selecting high-potential firms, internal records show that the committee also sought balanced representation across industries and regions and was further constrained by limited exhibition space and the high transport costs of exhibits. As a result, not all applicants could attend, and some high-potential firms were excluded in favor of others chosen to meet diversity and logistical requirements.

We exploit two features of this setting for identification. First, we use the prizes awarded in the trial exhibition to obtain an objective, ex-ante, binned measure of firms’ export potential. Second, we leverage the government’s constraints, which prevented perfect sorting and resulted in treated and untreated firms with similar export potential bins.

To address selection bias and estimate a causal treatment effect from attending the 1900 Paris Exhibition, we develop a novel estimation strategy that constructs “causal bands” for the treatment effect by creating two distinct control groups for treated firms, yielding upper and lower bound estimates. The upper bound compares selected firms to untreated firms *within* the same prize category, which may remain upward biased due to residual selection. The lower bound “stacks the deck” against finding a treatment effect by using a control group from the prize category immediately above the treated firms, providing a conservative estimate. This strategy also lends itself to a validation test: the upper bound estimate should exceed the lower bound, thereby confirming the assumptions about the selection process. Our estimation strategy improves upon alternative approaches such as matching estimators by exploiting additional information about whether potential controls are “close from above” or “close from below” in export potential rather than relying solely on “close” matches.

We build a novel panel dataset of Hungarian firms from five historical sources for the period between 1896 to 1905. We digitized and systematically processed over 12,000 firm records using automated text extraction, geographic standardization algorithms, and multi-criteria entity resolution to construct the first comprehensive panel of Hungarian industrial firms during this period. Our dataset contains information on export status, employment, mechanization (horsepower), and other firm-specific variables for approximately 3,700 Hungarian manufacturing firms tracked from 1896 to 1905, although not all variables are available in every year.

We validate our estimation strategy by examining pre-treatment characteristics for both the upper and lower bound approaches. As expected, in the upper bound estimation, treated firms slightly outperform untreated firms in several baseline characteristics. In contrast, in the lower bound estimation, there is no significant difference between treated and untreated firms; if anything, treated firms perform slightly worse at baseline.

For our main outcome of interest—exporting—we have two pre-exhibition observations, which

allow us to test for pre-trends. Consistent with expectations, the upper bound estimation shows a pre-trend, suggesting that it may be upward biased. In contrast, the lower bound estimation exhibits no pre-trend, supporting its interpretation as a more conservative estimate. Overall, these results suggest that the true causal effect likely lies closer to the lower bound, though within the estimated range.

We find that exhibition participation in Paris increased the export probability 4 to 5 years after the exhibition. For the broader export capability status, the causal band corresponds to a 8.5 to 14 percentage point higher probability in 1904. For the narrower definition of extra AHM export status, the band reflects a 4 to 10 percentage point higher probability in 1905. Beyond exports, we document substantial increases in employment among Paris exhibitors, with estimated gains of 15–23% over an eight-year period, equivalent to roughly 1.8–2.6 percentage points of annual growth. However, these employment gains are not accompanied by increases in mechanization. We find no statistically significant change in machine capacity, suggesting that firms expanded primarily through labor rather than capital-intensive upgrades.

At the same time, this average effect masks substantial heterogeneity that sheds light on search and matching frictions. We find that treatment effects are significantly smaller when Hungarian firms faced more competitors in their product category at the trade fair, consistent with congestion effects. Furthermore, firms in categories with more potential buyers will benefit the more than those in thinner markets.

Our work contributes to four strands of literature. First, we relate to a growing literature studying firm-to-firm relationships. The empirical literature has focused on showing that exogenous reductions in search and matching frictions cause more matches. Part of the literature has used quasi-experiments that reduce transportation and communication cost to show evidence of search and matching frictions (e.g., Jensen (2007); Jensen and Miller (2018); Allen (2014); Bernard et al. (2019)). Others employed RCTs to measure the effect of giving firms access to new suppliers (Cai et al., 2024) or buyers (Atkin et al., 2017; Bergquist et al., 2024).

A complementary literature has been developed that include searching and matching frictions or fixed cost associated with establishing new links to explain common patterns of supplier-buyer links, such as searching and matching frictions, or fixed cost associated with establishing new links (e.g., Bernard et al. 2019; Demir et al. 2024; Arkolakis et al. 2025; Eaton et al. 2022a), Dhyne et al. 2023; Lim 2017; Bernard et al. 2018, Eaton et al. 2022b; Miyauchi 2024). For international trade, Lenoir et al. (2023) structurally estimated search frictions using French firm-level data, finding them as important as traditional iceberg costs in impeding trade. Krolkowski and McCallum (2021) embedded goods-market frictions in a general equilibrium framework and show that matching frictions alter the gains from trade by creating an endogenous fraction of unmatched exporters.

We contribute to this literature by studying trade fairs—the most common mechanism firms use to form new links—which, unlike targeted interventions, simultaneously expose firms to potential partners and competitors, and unlike infrastructure improvements, operate in a selective, focused

environment for matching. Our findings on the heterogeneous effects of trade fairs across product categories with different levels of competition and buyer concentration speak directly to these mechanisms, providing causal evidence on how search and matching frictions operate when firms are brought together in a concentrated environment.

Second, we contribute to the literature evaluating export promotion programs (e.g., Munch and Schaur 2018; Martincus and Carballo 2010; Carballo and Volpe Martincus 2008; Broocks and Van Biesebroeck 2017; Munch et al. forthcoming; Van Biesebroeck et al. 2015; Hiller 2012; Görg et al. 2008; Schminke and Van Biesebroeck 2013). While early studies often relied on difference-in-differences matching based on observable firm characteristics, more recent work exploits quasi-random variation in the selection processes of export promotion agencies to identify causal effects. We extend this literature in three ways. First, we introduce a novel estimation approach that establishes “causal bands” for treatment effects, improving upon prior matching estimators by distinguishing between comparisons “close from above” and “close from below.” Second, whereas export promotion agencies sometimes subsidize trade fair participation as part of a broader package of services, we focus exclusively on trade fairs as a standalone intervention. Third, our setting allows us to examine not only the average effect on treated firms but also the role of competition among participants, highlighting how exposure to competitors can generate congestion effects.

Third, our work contributes to the growing body of quantitative research on how industrial policy influenced development during the first era of globalization. While the historical literature has documented the prevalence of industrial policy (see, e.g., the survey in (Juhász and Steinwender, 2024)), quantitative evaluations remain scarce. Notable exceptions include studies on temporary protection (Juhász, 2018; Hanlon, 2020), infrastructure improvements (Steinwender, 2018), intellectual property protection (Alfaro et al., 2022), colonialism (Romero, 2023), and tariffs (Juhász, 2018). In contrast, the role of trade fairs in shaping firm outcomes remains unexplored. Prior research has instead used trade fair exhibitor data as an alternative measure of innovation to examine questions related to intellectual property protection (Moser, 2005, 2012). This paper deals with a very planned and large scale drive yielding Hungary becoming the 5th largest exhibitor in 1900 vs being of the size of Vienna in 1878.

Finally, our estimation approach builds on the literature using bounds to address selection problems in program evaluation, deriving credible treatment effect bounds under weaker assumptions (e.g., Manski 2003; Lee 2009; Manski and Pepper 2000; Tamer 2010; Ho and Rosen 2017; Blundell et al. 2007). Our setting shares key features with this literature: we face a selection problem where exhibiting firms likely differ from non-exhibitors in unobservable ways. Like Manski and Pepper (2000), we exploit an ex-ante ranking variable—the Budapest exhibition awards—that partially predicts treatment assignment. However, our approach innovates in how we construct bounds. Rather than using trimming procedures as in Lee (2009), or comparing across levels of the monotone treatment selection pattern as in Manski and Pepper (2000), we exploit the discrete nature of our award categories to construct two distinct control groups (same vs. lower-tier awards). This

approach offers several advantages: it provides clear economic intuition by comparing treated firms to those of similar versus superior export potential and incorporates a built-in validation test, as the upper bound should exceed the lower bound. We’ll show comparisons of our method versus the Manski bands.

Section 2 provides the context for the historical experiment. Section 3 outlines a novel empirical strategy for constructing “causal bands” for the treatment effect. Section 4 describes the construction of a panel dataset of Hungarian firms by digitizing and linking multiple historical sources. Section 5 details the application of this empirical strategy to the setting, and Section 6 presents the results. We then look at heterogeneity of treatment effects across product subcategories to understand the role of competition and market thickness in Section 7. We conclude in Section 8 and relate results to contemporary debates.

2 Hungarian firms in the 1900 Paris World Exhibition

The 1900 Paris *Exposition Universelle* was conceived as the largest world’s fair to date, celebrating the achievements of the 19th century and showcasing visions for the 20th. Running from April to November 1900, it attracted over 48 million visitors—roughly equivalent to France’s entire population. The exhibition featured 76,000 exhibitors displaying innovations ranging from escalators and talking films to diesel engines and telegraphs. Exhibitors were divided into 18 categories (reflecting broad industries) based on the general theme of their products, and further subdivided into 121 subcategories (reflecting narrower industries) specific to their exhibits.

The exhibition took place in halls at different parts of Paris with great many by the *Champs de Mars*. Companies in the same broad category shared an exhibition hall. For instances in the bottom left of the group of halls had “Mines et métallurgie” (mining and metallurgy). Within that, firms were by country, and Hungary was beside Austria and next to Belgium.

For participating countries, the exhibition served multiple purposes. Beyond cultural diplomacy and national prestige, governments explicitly recognized its commercial potential. The vast concentration of international visitors—including merchants, industrialists, and potential customers—created unprecedented opportunities for establishing business connections (Tamilia, 2007). Contemporary accounts describe the exhibition grounds as bustling with commercial negotiations alongside the display of innovations.

The exhibition’s commercial importance was amplified by the communication constraints of the era. International business relationships typically required either costly travel or slow correspondence by mail. The telegraph, while revolutionary for time-sensitive information, was most effective for trade in homogeneous products that could be codified, but limited in facilitating trade in goods that required physical inspection (Juhász and Steinwender, 2018). In this context, the exhibition offered a unique opportunity for face-to-face meetings that could build trust and enable detailed product demonstrations.

By 1900, Hungary was undergoing rapid industrialization within the dual monarchy of Austria-Hungary. While still predominantly agricultural, the country had developed notable manufacturing capacity, particularly in food processing (especially flour milling), textiles, and machinery (Klein et al., 2017). The government actively supported industrial development through policies such as subsidies, infrastructure investment, and export promotion (Kun, 2019).

The Hungarian government’s approach to the Paris Exhibition reflected careful strategic planning. Since 1887, institutional efforts to prepare and manage Hungarian participation in international trade fairs had been prioritized to promote nascent industries. The 1896 Budapest Millennial Exhibition, initially planned as a world fair, served as a rehearsal, allowing officials to identify the country’s most competitive products and firms (Székely, 2015). Subsequently, for the selected exhibitors, the government covered all participation costs for the selected exhibitors. Beyond national prestige, the goal of establishing international business connections was explicitly emphasized by government officials, who noted that Paris provided “a favorable opportunity for establishing commercial connections” to promote Hungarian exports.¹

3 Empirical Strategy

3.1 The Selection Challenge

The central challenge in estimating the causal effect of trade fair participation is selection: exhibiting firms are likely to differ systematically from non-exhibitors. Let $Y_i(1)$ and $Y_i(0)$ denote the potential outcomes for firm i with and without Paris exhibition participation, respectively. The firm-level treatment effect is:

$$\tau_i = Y_i(1) - Y_i(0).$$

Since only one potential outcome is observed, realized outcomes are:

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0),$$

where D_i is an indicator for participation. A naive comparison of participants and non-participants recovers:

$$\begin{aligned} E[Y(1)|D = 1] - E[Y(0)|D = 0] &= \\ &= E[Y(1)|D = 1] - E[Y(0)|D = 1] + E[Y(0)|D = 1] - E[Y(0)|D = 0] \\ &= \underbrace{E[\tau|D = 1]}_{\text{ATT}} + \underbrace{E[Y(0)|D = 1] - E[Y(0)|D = 0]}_{\text{selection bias}} \end{aligned}$$

If firms with greater export potential are more likely to participate, the selection term $E[Y(0)|D =$

¹Hungarian Industrial Journal 1897 ed. 2, by Baron Ernő Dániel, Hungarian Royal Minister of Commerce

$1] - E[Y(0)|D = 0]$ is positive, biasing the naive difference upward relative to the average treatment effect on the treated (ATT).

We address this challenge by exploiting a preceding trial exhibition that provides an objective, ex-ante categorical measure of firms’ “export potential.” We use this measure to construct two distinct control groups for the treatment group, yielding upper and lower bound estimates of the causal effect. The upper bound compares participants to a control group of similar export potential but may still be upward biased due to residual selection. The lower bound “stacks the deck” against finding an effect by using a control group with even higher export potential, thereby providing a conservative estimate.

3.2 The role of the Budapest trial exhibition

Our ex-ante measure of export potential is derived from the 1896 Budapest Millennial Exhibition, organized as a “trial exhibition” four years prior to the Paris Exhibition. The exhibition was widely advertised in hundreds of newspapers and thousands of posters, ensuring broad recognition and resulting in a large fair with over 9,000 exhibitors. An independent jury assessed each exhibit’s export potential and awarded four prizes in descending order: gold, silver, bronze, and honorable mention.² We classify firms into five categories—four award levels plus a residual “no award” group—denoted $A_i \in \{1, 2, 3, 4, 5\}$, where $A_i = 1$ indicates the highest export potential. Formally, we impose the following assumption:

Assumption 1 (Export potential bins). *On average, firms in higher award categories have (weakly) higher export potential:*

$$E[Y(0) \mid A_i = a] \geq E[Y(0) \mid A_i = a + 1].$$

Assumption 1 ensures that the Budapest awards provide an ordered proxy for untreated export performance. We use this to interpret the award categories as discrete export-potential bins and to motivate the selection assumptions below. Given this, we will then impose two additional assumptions to construct the bounds below.

3.3 Institutional context on selection for Paris

After the trial exhibition, a separate selection committee—distinct from the 1896 jury—was tasked with choosing firms for the Paris Exhibition. While the main goal was to promote Hungarian exports by selecting firms with high export potential, internal documents show that the committee also aimed for balanced representation across industries and regions.

In addition to export potential, the committee emphasized industrial (and regional) diversity, seeking to present a broad cross-section of Hungarian production. There was also a constraint:

²For simplicity, we relabeled the names, the original names and descriptions of the awards are found in Appendix ??; more on criteria and justification for awards in Appendix ??; more on the jury selection in Appendix ??

exhibition space in Paris was limited, transport costs were high, and thus not all applicants could be accommodated. Within some product categories, space constraints meant that even firms with export potential were excluded, while some firms would be included to maintain sectoral balance.

Figure 1 displays the deviations between each industry’s share of firms receiving gold or silver awards and their overall share among Budapest exhibitors. A higher positive deviation—especially notable for the food industry—indicates a particularly successful sector, where the proportion of high awards exceeds the industry’s overall representation. The second set of deviations compares the share of firms selected for the Paris exhibition to their share in the Budapest exhibition, highlighting over- or under-representation by industry. While the food industry remained overrepresented in Paris, the degree of overrepresentation was notably smaller than what would be expected based on its strong performance in the Budapest awards. This indicates that the selection committee balanced firms across industries, thus not only firms with the highest export potential went to Paris.

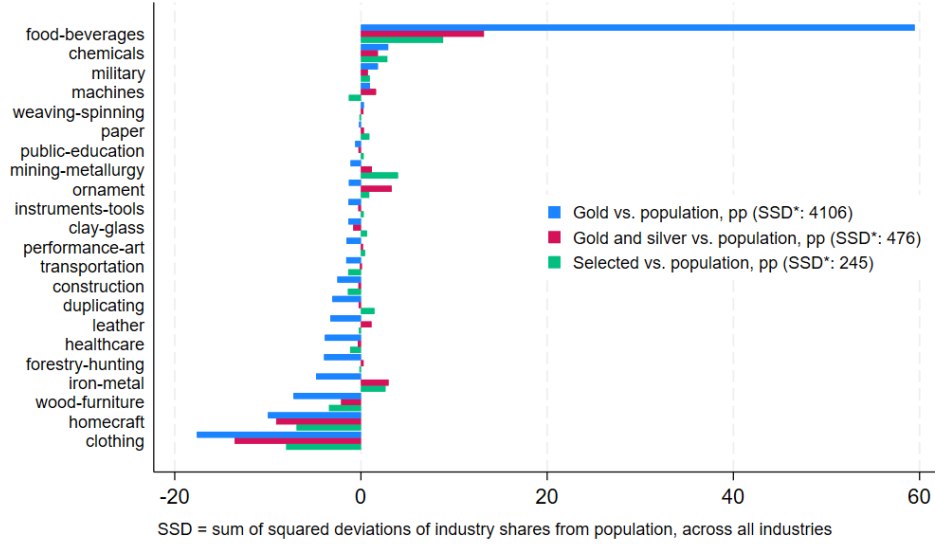


Figure 1: Paris exhibitors are more balanced across industries than gold or silver awards

3.4 Modeling selection probabilities

Let $\pi_a = P(D_i = 1 \mid A_i = a)$ denote the selection probability for award category a . Table 1 illustrates this selection process. Overall, 15% of firms were selected for Paris. Participation is strictly positive in all award categories ($\pi_a \in (0, 1)$ for all a) and declines monotonically with award rank, from 60% in the top category to 7% in the lowest: $\pi_1 > \pi_2 > \dots > \pi_5$. This pattern is consistent with our interpretation of the awards as ranking export potential, Assumption 1, and with a selection committee that prioritizes firms with higher export potential.

Table 1: Share of Paris participants, by 1896 award category

Award category	Number of firms	Share of firms	
		Paris exhibitors	Non-Paris exhibitors
Gold medal	140	60%	40%
Silver medal	425	36%	64%
Bronze medal	1,034	16%	84%
Honorable mention	1,136	7%	93%
No award	928	7%	93%
Total	3,733	552	3,181
Total (%)	100%	15%	85%

3.5 Bounding the Treatment Effect

We leverage this structure to define our bounding estimates, by comparing the treatment group to differing control groups.

3.5.1 Upper Bound

We saw that the selection probability declines with award rank across categories. It is therefore plausible that even within categories, firms with higher export potential were more likely to be selected, after consideration of the constraints. We formalize this in the following assumption, which underpins our construction of a conservative upper bound:

Assumption 2 (Upper bound). *Within award categories, selected firms have (weakly) higher export potential than non-selected firms:*

$$E[Y_i(0) \mid D_i = 1, A_i = a] \geq E[Y_i(0) \mid D_i = 0, A_i = a], \quad \forall a.$$

This assumption implies residual selection bias within award categories, leading to upward-biased estimates of the treatment effect for each award category.

Lemma 1 (Upper bound). *Define the upper bound estimate for each award group a as $\beta_{upper}^a := E[Y(1) \mid D = 1, A = a] - E[Y(0) \mid D = 0, A = a]$. Define the average treatment effect on the treated for each award category as $\tau^a := E[Y(1) \mid D = 1, A = a] - E[Y(0) \mid D = 1, A = a]$. Under assumption 2,*

$$\beta_{upper}^a \leq \tau^a,$$

i.e., β_{upper}^a is an upper bound for the treatment effect on the treated.

Proof.

$$\begin{aligned}
\beta_{upper}^a &= E[Y(1) \mid D = 1, A = a] - E[Y(0) \mid D = 0, A = a] \\
&= E[Y(1) \mid D = 1, A = a] - E[Y(0) \mid D = 1, A = a] \\
&\quad + E[Y(0) \mid D = 1, A = a] - E[Y(0) \mid D = 0, A = a] \\
&= \tau^a + \underbrace{E[Y(0) \mid D = 1, A = a] - E[Y(0) \mid D = 0, A = a]}_{\geq 0}.
\end{aligned} \tag{1}$$

□

We then pool observations across all award categories to estimate the overall upward-biased treatment effect using the following regression specification:

$$\Delta Y_i = \beta_{upper} \cdot D_i + \sum_a \gamma_a \cdot \mathbf{1}\{A_i = a\} + X_i' \delta + \epsilon_i, \tag{2}$$

where ΔY_i denotes the change in the outcome, $\mathbf{1}A_i = a$ are award category fixed effects that ensure within-category comparisons, and X_i includes baseline controls, such as the initial level of the outcome.

3.5.2 Lower Bound

To construct the lower bound, we “stack the deck” against finding a treatment effect by using a control group with higher export potential than the treated group. Specifically, we compare treated firms from each award category to untreated firms from the category immediately above. Formally, we impose the following assumption, which is a stronger assumption than Assumption 1.

Assumption 3 (Lower bound). *Non-selected firms from each award category have (weakly) higher export potential than selected firms from the category below:*

$$E[Y_i(0) \mid D_i = 0, A_i = a - 1] \geq E[Y_i(0) \mid D_i = 1, A_i = a].$$

Lemma 2 (Lower bound). *Define the lower bound estimate for each award group a as $\beta_{lower}^a := E[Y(1) \mid D = 1, A = a] - E[Y(0) \mid D = 0, A = a - 1]$. Under assumption 3,*

$$\beta_{lower}^a \geq \tau^a,$$

i.e., β_{lower}^a is a lower bound for the treatment effect on the treated.

Proof.

$$\begin{aligned}
\beta_{lower}^a &= E[Y(1)|D = 1, A = a] - E[Y(0)|D = 0, A = a - 1] \\
&= E[Y(1)|D = 1, A = a] - E[Y(0)|D = 1, A = a] \\
&\quad + E[Y(0)|D = 1, A = a] - E[Y(0)|D = 0, A = a - 1] \\
&= \tau^a + \underbrace{E[Y(0)|D = 1, A = a] - E[Y(0)|D = 0, A = a - 1]}_{\leq 0}
\end{aligned}$$

□

We implement this by creating comparison groups $g \in \{1, 2, 3, 4\}$, where group g contains treated firms in award category $g + 1$ and untreated firms in award category g . Comparing treated and untreated firms within the same comparison group g yields downward-biased estimates for each comparison group g .

We aggregate these comparison-group-specific downwards-biased treatment effects across comparison groups using the following regression:

$$\Delta Y_i = \beta_{lower} \cdot D_i + \sum_g \theta_g \cdot \mathbf{1}\{G_i = g\} + X_i' \delta + \epsilon_i, \quad (3)$$

where G_i denotes comparison group membership. Note that this specification changes the sample by excluding treated firms in the highest award category (as no higher-ranked control group exists) and untreated firms in the lowest award category (as no lower-ranked treatment group exists).

3.5.3 Validation Test

Our setting lends itself to the following validation test:

Lemma 3 (Validation test). *Under Assumptions 2 and 3, it follows that*

$$\beta_{upper} \geq \beta_{lower}.$$

If this inequality does not hold in the data, at least one of Assumptions 2 or 3 is rejected. However, this constitutes only a necessary, not a sufficient, condition for the validity of the estimation strategy: if the inequality holds, the assumptions may or may not hold.

3.6 Benefits of causal bounds

Our estimation strategy improves upon alternative approaches to identifying causal treatment effects in this context. For example, matching estimators (e.g., coarsened exact matching based on awards; propensity scores based on awards; nearest neighbor matching based on awards) would all

compare treated and untreated firms within the same award category. As shown in equation (1), such within-award category comparisons are likely subject to residual selection bias of unknown but plausibly positive sign (as in our Assumption 2). The intuition behind our approach is that, rather than relying solely on "close" matches, we exploit additional information about whether potential controls are "close from above" or "close from below" in export potential and use this to construct bounds on the treatment effect.

The award categories can be viewed as a binned running variable in a fuzzy RDD. However, because the binning masks selection within categories, the standard fuzzy RDD approach would be invalid.

3.7 Relation to the Partial Identification Literature

Our approach of constructing causal bands relates to a broader econometrics literature on partial identification and bounding treatment effects when point identification is not feasible. The classic contributions include Lee (2009) who developed trimming bounds for sample selection, and Manski (2003) who provides a general framework for bounding treatment effects under weak assumptions. More recently, this literature has expanded significantly, as surveyed by Tamer (2010) and Ho and Rosen (2017).

In particular, our approach builds on the literature using bounds to address selection problems in program evaluation deriving credible bounds on treatment effects under weaker, more plausible assumptions. A foundational challenge in this literature is sample selection, where outcomes are observed only for a selected subsample. Lee (2009) addresses this problem in the context of job training programs, where wages are observed only for employed individuals. Since training may affect both employment and wages, simple comparisons of wages among the employed suffer from selection bias. Lee uses a strict monotonicity assumption—that training cannot hurt employment prospects—and develops a trimming procedure that yields sharp bounds on the average treatment effect. Blundell et al. (2007) tackle a related problem when studying wage inequality over time with changing employment rates. They bound the wage distribution at unobserved quantiles by combining the observed distribution for employed individuals with worst-case assumptions about wages of the non-employed.

Another relevant methodology is Manski and Pepper (2000) who introduce a method of partial identification through monotone instrumental variables (MIV). In their application to returns to schooling, they observe that while AFQT scores do not perfectly determine educational attainment, they provide a credible ranking: individuals with higher test scores are more likely to obtain more schooling. This monotone treatment selection (MTS) pattern—combined with assumptions about monotone treatment response—allows them to bound the returns to schooling without requiring a valid exclusion restriction. The key insight is that comparing outcomes across different levels of the ranking variable provides information about treatment effects even when the ranking does not fully determine treatment.

Our setting shares fundamental features with this literature: we face a selection problem where exhibiting firms likely differ from non-exhibitors in unobservable ways. Like Manski and Pepper (2000), we have an ex-ante ranking variable—the Budapest exhibition awards—that partially predicts treatment assignment. The monotonic decline in Paris selection probabilities across award categories (from 60% to 7%) mirrors their monotone treatment selection pattern.

However, our approach innovates in how we construct bounds. Rather than using trimming procedures as in Lee (2009), or comparing across levels of the MIV as in Manski and Pepper (2000), we exploit the discrete nature of our award categories to construct two distinct control groups (same vs lower tier awards). This approach offers several advantages. First, it provides transparent economic intuition: we bound the treatment effect by comparing to firms of similar versus superior export potential. Second, it yields a testable implication—that the upper bound should exceed the lower bound—providing a validation check absent in sharp bounds approaches. Third, it leverages unique institutional features of our setting, where the trial exhibition provides an objective, pre-treatment measure of firm quality that is rarely available in modern applications. One important disadvantage is that our bounds may not be sharp in the sense of being the narrowest possible given our assumptions. At the same time they provide credible and interpretable limits on the causal effect.

Recent applications combine policy evaluation with bounds estimation in various contexts.³ Assunção et al. (2023) use the changes-in-changes framework of Athey and Imbens (2006) to evaluate deforestation policies in the Amazon, obtaining bounds when support conditions are violated. Rambachan and Roth (2023) develop methods for constructing bounds in difference-in-differences settings when parallel trends assumptions may be violated. In the regression discontinuity context, Gerard et al. (2020) develop bounds for regression discontinuity designs when agents can manipulate the running variable through costly effort, causing selection around the threshold. Our setting has analogous features: if firms could influence their Budapest awards or Paris selection through bribery, this would constitute costly manipulation. This aspect reinforces the need for the lower bound.

4 Data

Table 2 provides an overview of the data sources used in this study. All data were originally obtained from printed sources and subsequently digitized. Data on firm characteristics are described in Békés et al. (2026).

The analysis required extensive digitization and text processing of historical archival sources housed in the Hungarian National Library.⁴

³Other use of bounds may be related to computation difficulties. Facing impossibly rich potential export decision, Morales et al. (2019) use moment condition inequalities to create bounds for export decisions at the firm level.

⁴All primary sources, accessed via microfilm, underwent systematic digitization involving photographic

Name	Year	Key Variables
Budapest Exhibition Catalog	1896	Exhibitor names, awards, product descriptions
Paris Exhibition Catalog	1900	Exhibitor names, awards, product descriptions, competitors
Hungarian Museum of Commerce Survey 1	1898	Firm characteristics; export readiness (outside region, within and beyond AHM)
Hungarian Museum of Commerce Survey 2	1904	Firm characteristics; export readiness (outside region, within and beyond AHM)
Austro-Hungarian Export Compass	1905	Firm characteristics; export activity outside AHM customs territory

Table 2: Overview of historical data sources

Firm names were identified across multiple datasets, with considerable effort devoted to ensuring unique identification. Indeed, the most complex technical challenge involved entity resolution across five independently collected datasets to construct a longitudinal panel. Using matching algorithms, we created correspondence tables, did a great deal of manual checking and consolidated them into a master identification system enabling both balanced and unbalanced panel specifications. Further details on the digitization process and entity resolution methods are provided in the Appendix B

4.1 Data from exhibitions

4.1.1 1896 Budapest

Our primary dataset is based on the catalog of the 1896 Budapest Millennial Exhibition, which documented 9,650 exhibitors from Hungary (excluding Croatia-Slavonia and Bosnia-Herzegovina). To focus on commercially oriented manufacturing firms, we excluded several categories: historical artifacts (880 exhibitors), fine arts (269), education (697), agriculture (3,175) and finance (344). After removing further non-commercial entities such as municipalities and NGOs, and consolidating duplicate entries where firms appeared across multiple categories, our refined sample comprises 3,733 for-profit manufacturing firms.

The exhibition catalog provides firm-level information, including company names, addresses, exhibited products, and industry categories for nearly all participants.

For a substantial subset of exhibitors, additional details are available: workforce composition capture, optical character recognition using Google Document AI, and systematic error correction through comparison with original images. The resulting raw OCR output was processed to standardize Hungarian diacritical marks and historical orthographic conventions, extract structured data fields from semi-structured catalog entries, and implement manual verification procedures to correct automated extraction errors. Geographic standardization employed string similarity algorithms to match historical firm addresses to the GISTA Hungarorum municipal database, enabling inference of administrative hierarchies from municipality to statistical region levels.

(skilled and unskilled workers), production capacity, output values in Hungarian Forints, export destinations, founding years, and patent holdings. Summary statistics are presented in Table 3. We matched these firms to the official awards list published in the Budapest Millennial Exhibition Brochure following the exhibition’s conclusion in 1896.

4.1.2 1900 Paris

We identify treated firms using the catalog from the 1900 Paris *Exposition Universelle*, classifying any firm from our baseline sample as a Paris exhibitor if we successfully match it to an entity listed in the Paris catalog.

To proxy competition, we extracted exhibitor counts by country and 52 subcategories from the General Catalog, and calculated the number of total exhibitors.

4.2 Outcomes

We derive post-treatment outcomes from two primary sources that provide complementary measures of export activity—export capability and export status—as well as other firm performance indicators. Firm level characteristics come from a panel data we created from several sources, see details in Békés et al. (2026). Below we only discuss the relevant aspects of the dataset.

4.2.1 Export capability

The first source is a set of comprehensive firm surveys conducted by the Hungarian National Museum of Commerce in 1898 (establishing pre-treatment baselines) and 1904 (measuring post-treatment outcomes). The stated objective of the survey, as noted in its documentation, was to compile a registry of firms that were either actively exporting or *ready to export* beyond the borders of the Hungarian Kingdom. Specifically, manufacturers that “only lacked a foreign order to become export companies” were included. The Museum defined export capability as encompassing “industrial plants that exceed local consumption demands, are prepared for international shipping, and maintain capacity to engage with foreign importers.” This also included firms selling outside their region (e.g., from Budapest to Slavonia [now part of Croatia] or Pozsony [now Bratislava, Slovakia]).⁵ For all firms, we construct an export capability dummy equal to 1 if a firm appears in these registries in a given year. In addition, the survey provides data on employment levels, albeit for only a subset of firms.

4.2.2 Export status

Our second outcome source is the 1905 Austro-Hungarian Export Compass, compiled by the Austrian Ministry of Commerce in partnership with regional Chambers of Commerce. This compre-

⁵For additional details, see Appendix ??.

hensive directory documents all firms that conducted export activities outside the Austria-Hungary customs union during the reference year, organized by product category.

For all firms, we construct an export status dummy equal to 1 if a firm appears in these directories in a given year. This source enables us to identify firms that successfully transitioned from domestic to international markets. This data has the advantage of measuring the starkest aspect of export – outside not only Hungary but the Austrian-Hungarian Monarchy.

Table 3: Summary statistics

Variable	N	Mean	SD	Min	Max
Exporter	3733	0.15	0.36	0	1
Worker info	3733	0.33	0.47	0	1
Horsepower info	3733	0.18	0.39	0	1
Output info	3733	0.19	0.39	0	1
Worker number	1022	172	675	1	12,968
Share of skilled	1008	0.75	0.30	0.01	1
Horsepower number	512	208	680	1	9206
Output (in HUF)	553	0.6 mil	1.7 mil	400	14 mil
Age (in yrs)	1418	26	33	1	510
Patent	3733	0.05	0.21	0	1
Multi-location firms	3733	0.06	0.24	0	1
Joint-stock company	3733	0.04	0.20	0	1

5 Estimating Framework

5.1 Estimated regression

We implement our bounding strategy through the following estimating equation:

$$\Delta y_i = \beta \text{exhibitor}_i + \text{BoundGroupFE}_j + FE_p + y_{i,pre} + \epsilon_i \quad (4)$$

where Δy_i represents the change in outcome for firm i , exhibitor_i is an indicator for Paris exhibition participation in 1900, BoundGroupFE_j denotes the fixed effects that define our comparison groups, FE_p represents product fixed effects, and $y_{i,pre}$ is the baseline value of the outcome variable.

The key to our identification strategy lies in how we construct the bound group fixed effects. For the upper bound estimation, we set BoundGroupFE_j as award category dummies, ensuring that we compare treated and untreated firms within the same Budapest award class. This specification uses the full sample of firms and exploits within-award variation. Since better firms within each award category were more likely to be selected for Paris, this comparison yields an upward-biased estimate of the treatment effect.

For the lower bound estimation, we construct BoundGroupFE_j as comparison group dummies that match untreated firms from higher award classes with treated firms from lower award classes. This requires dropping treated firms in the top award category (who have no higher-quality control group) and untreated firms in the lowest award category (who have no lower-quality treatment group). Since the control firms in this specification have systematically higher quality than the treated firms, we obtain a downward-biased estimate.

5.2 Comparing bounds

In this section, we validate our estimation strategy by examining pre-treatment characteristics and pre-trends for both the upper and lower bound approaches.

5.2.1 Balance on Pre-Treatment Characteristics

Figure 2 presents coefficient plots from regressions of treatment indicators on baseline firm characteristics in 1896, varying both the included group fixed effects and the sample used for the lower and upper bound estimates, as described in Section 5.1. The left panel displays the upper bound estimation and shows that, within award categories, treated firms were more likely to report employment and machine capacity, and to be joint-stock companies, multi-location firms, or exporters. These differences align with our expectation that selection favored higher-quality firms within categories, contributing to the upward bias in the upper bound estimates.

By contrast, the right panel displays the lower bound estimation. Here, treated firms exhibit no significant advantages in observable characteristics relative to their higher-award control group. The point estimates are generally closer to zero or even negative, consistent with our design of comparing treated firms with lower export potential to control firms with higher export potential. This pattern supports our interpretation that the lower bound estimates are conservative.

Figure 2: Balance of covariates for dummy variables

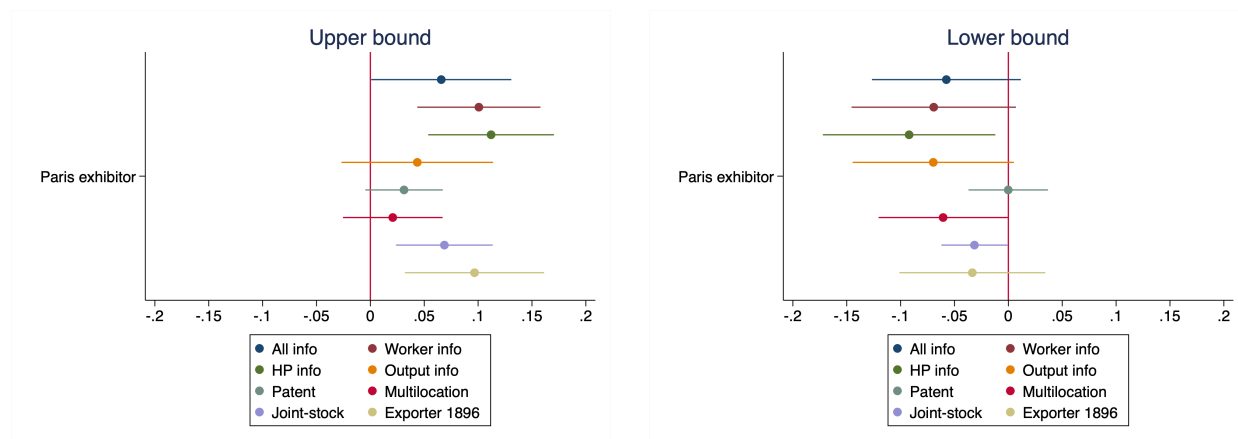
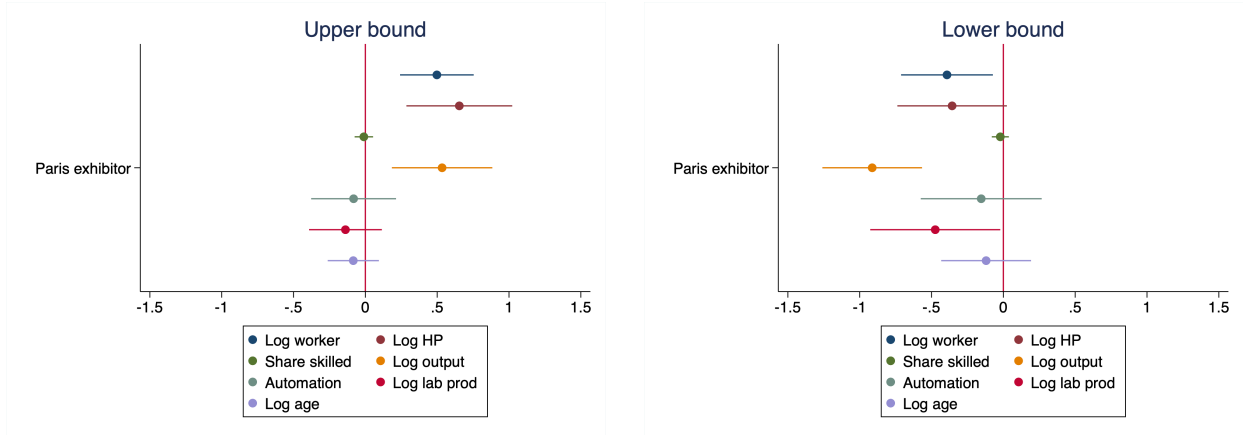


Figure 3: Balance of covariates for continuous variables



A similar pattern is observed in Figure 3, which shows the balance of numerical pre-treatment covariates. Within award categories, treated firms tend to be larger both in terms of employment, machine capacity, and output. Labor productivity, firm age, and input share, however, are balanced. By using the lower sample with treatment groups from lower classes, the size imbalances are corrected or even reversed.

5.2.2 Pre-Trends

For our main outcome of interest—export status—we have two pre-exhibition observations, 1896 and 1898, which allow us to test for pre-trends. Table 4 implements both the upper and lower bound estimation as described in section 5.1, using the change in export status as outcome.

Table 4: Pre-Trends Analysis: Change in Export Status, 1896-1898

	(1)	(2)	(3)	(4)
	Upper bound		Lower bound	
Paris exhibitor	0.149*** (0.0262)	0.121*** (0.0219)	0.118*** (0.0268)	0.0129 (0.0303)
Export status 1896	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	Yes	Yes
Bound group dummies	No	Upper	Upper	Lower
Observations	3,733	3,733	2,721	2,721
R-squared	0.018	0.291	0.293	0.289

Notes: Sample of firms: columns (1) and (2) include all firms from the Budapest exhibition. Columns (3) and (4) only include firms used in the lower bound estimation sample. Standard errors are clustered by product. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The naive OLS estimate in column (1) regresses the change in export status only on a treatment dummy and lagged exporter status, without including any fixed effects. It shows that future Paris exhibitors were already experiencing faster export growth prior to the exhibition, with a 13.9 percentage point higher probability of starting to export between 1896 and 1898. Column (2) implements our upper bound estimation strategy by adding award category and product fixed effects. This within-award category comparison reduces the estimated selection bias slightly, to 12.1 percentage points. The remaining pre-trend confirms that even the upper bound estimates may be upward biased.

Column (4) presents our lower bound estimation, which differs from column (2) in two steps that are separated in column (3). First, column (3) restricts the sample to firms used in the lower bound estimation (i.e., dropping treated firms in the top award category and untreated firms in the lowest award category) but retains the fixed effects. This adjustment alone is insufficient to eliminate pre-trends. Column (4) then adds comparison group dummies that match treated firms in lower award categories to untreated firms in higher award categories. Crucially, this lower bound specification shows no significant pre-trend: the coefficient falls to 1.77 percentage points and loses statistical significance.

The disappearance of pre-trends when relying on the higher-quality comparison group suggests that the causal effect is likely closer to the lower bound estimates.

6 The effect of the Paris exhibition

This section presents our estimates of the causal effect of exhibiting in Paris in 1900 on export capability, export status, employment growth, and technological upgrading. We also examine a key mechanism related to search and matching—the scope of competition from peer exhibitors. All results are presented using the causal bands described above.

6.1 Exports

6.1.1 Export capability

Table 5 reports the causal band estimates using the change in export capability as outcome.⁶ As in Table 4, which tested for pre-trends, we begin in column (1) with a naive OLS estimate that regresses the change in export capability on a treatment dummy and lagged exporter status, without including any fixed effects. Firms that exhibited in Paris are 19.5 percentage points more likely to have export capability in 1904. Column (2) implements our upper bound estimation strategy by adding award category and product fixed effects. This within-award category comparison reduces the estimated effect slightly, to 13.8 percentage points.

⁶Change in export capability is defined as the difference between export capability in 1904 and 1898.

Column (4) presents our lower bound estimation, which reduces the estimated effect to 8.5 percentage points, though it remains statistically significant. The fact that even our most conservative lower bound is positive and significant provides causal evidence of the impact of trade fair participation on export capability. Note that baseline export frequency is 12% and thus, the effect is very large in magnitude.

As in Table 4, column (3) provides an intermediate step between the upper and lower bound estimations by restricting the sample to firms included in the lower bound estimation while retaining the fixed effects used for the upper bound. Because the coefficient in column (3) is nearly identical to that in column (2), we confirm that the main driver of the lower bound estimate is the change in the control group induced by the modified fixed effects structure.

Table 5: Treatment Effect on Change in Export Capability, 1904-1898

	(1)	(2)	(3)	(4)
		Upper bound		Lower bound
Paris exhibitor	0.193*** (0.0270)	0.139*** (0.0228)	0.141*** (0.0230)	0.0863*** (0.0255)
Export status 1898	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	Yes	Yes
Bound group dummies	No	Upper	Upper	Lower
Observations	3,733	3,733	2,721	2,721
R-squared	0.254	0.323	0.325	0.325

Notes: Sample of firms: columns (1) and (2) include all firms from the Budapest exhibition. Columns (3) and (4) only include firms used in the lower bound estimation sample. Standard errors are clustered by product. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.1.2 Export status

While it is encouraging that the trade fair increased export capability, our ultimate interest lies in actual exporting. To test this, we examine the change in export status as the outcome in Table 6, retaining the same column structure as in Table 5.

Table 6 confirms positive and significant treatment effects on export status: the upper bound estimate is 11.6 percentage points (column 2), while the lower bound estimate remains positive at 4.1 percentage points (column 4). As before, a naive regression without controls substantially overestimates the effect (column 1), and the shift in control group—rather than the change in sample—drives the lower bound estimate.

Table 6: Treatment Effect on Change in Export Status, 1905-1896

	(1)	(2)	(3)	(4)
		Upper bound		Lower bound
Paris exhibitor	0.173*** (0.0273)	0.102*** (0.0216)	0.117*** (0.0235)	0.0432* (0.0237)
Export status 1896	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	Yes	Yes
Bound group dummies	No	Upper	Upper	Lower
Observations	3,733	3,733	2,721	2,721
R-squared	0.403	0.482	0.491	0.489

Notes: Sample of firms: columns (1) and (2) include all firms from the Budapest exhibition. Columns (3) and (4) only include firms used in the lower bound estimation sample. Standard errors are clustered by product. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.2 Comparison to Manski bounds

As noted earlier, our method is related to Manski bounds (Manski, 2003). We can benchmark our result to the original bounds as well as those disciplined by assumptions: First, the *Monotone Treatment Response (MTR)* assumption posits that $Y(1) \geq Y(0)$ for all units. Second, the *Monotone Treatment Selection (MTS)* assumption posits that treated units have weakly higher (or lower) potential outcomes than untreated units. All details are explained in the Appendix A.

Figure A.1 shows the comparison of our bounds and those created with the Manski method. We plotted the progression with MTR and MTS. From Manski bounds of $[-0.21, 0.79]$, MTR yields $[0.00, 0.79]$ (length ≈ 0.79), MTS yields $[-0.21, 0.39]$ (length ≈ 0.60), and their intersection gives $[0.00, 0.39]$ (length ≈ 0.39). Our award-based bounds of $[0.085, 0.14]$ (length ≈ 0.055) provide much sharper inference.

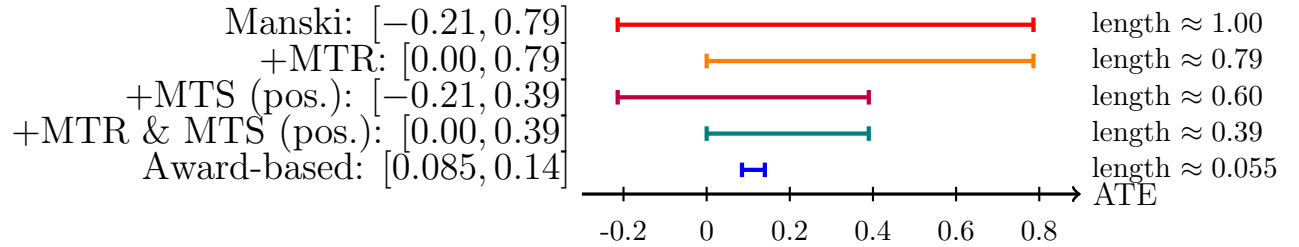


Figure 4: Bounds on ATE for Exporter Status in 1904 with MTR and MTS

6.3 Employment growth

Table 7 examines whether the trade fair also led to subsequent employment growth, using the change in log of workers as outcome. Our estimates range from 15.6% (lower bound, column 4) to 23.2% (upper bound, column 2), and remain statistically significant despite the smaller sample size driven by the availability of employment information. An employment increase of 15–23% over eight years corresponds to an annual growth rate of 1.8–2.6%. For a median firm with 35 workers in 1896, this translates into the creation of approximately 5–8 additional jobs by 1904.

Table 7: Treatment Effect on Employment Growth, 1896-1904

	(1)	(2)	(3)	(4)
		Upper bound		Lower bound
Paris exhibitor	0.246*** (0.055)	0.229*** (0.0625)	0.173*** (0.0760)	0.161* (0.0825)
ln(workers), 1896	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	Yes	Yes
Bound group dummies	No	Upper	Upper	Lower
Observations	624	624	518	518
R-squared	0.092	0.183	0.165	0.163

Notes: Sample of firms: columns (1) and (2) include all firms from the Budapest exhibition where the number of workers is observable both in 1896 and 1904. Columns (3) and (4) only include firms used in the lower bound estimation sample. Standard errors are clustered by product. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.4 Mechanization

We also examine whether firms upgraded their technology through increased mechanization, measured by the horsepower of installed machinery. Table 8 reports these results using the change in the log of horsepower for the sample with available data.

In contrast to the strong employment effects, we find no meaningful impact on mechanization. The upper bound estimates suggest at most a 10% increase in horsepower, but the effect is statistically insignificant. The lower bound estimates are essentially zero.

This pattern—employment growth without mechanization—indicates that firms expanded primarily through labor-intensive methods rather than capital deepening or technological change. Our findings contrast with Atkin et al. (2017), who show that providing individual firms with export orders to high-income countries leads to quality upgrading through learning. This suggests that trade fairs operate through a different channel for knowledge spillovers, potentially influenced by the presence of competitors.

Table 8: Treatment Effect on Mechanization (Change in log Horse Power), 1896-1904

	(1)	(2)	(3)	(4)
		Upper bound		Lower bound
Paris exhibitor	0.0459 (0.0855)	0.111 (0.997)	0.0434 (0.118)	0.0253 (0.0103)
ln(horse power), 1896	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	Yes	Yes
Bound group dummies	No	Upper	Upper	Lower
Observations	404	404	320	320
R-squared	0.058	0.198	0.216	0.219

Notes: Sample of firms: columns (1) and (2) include all firms from the Budapest exhibition where machine capacity in horsepower is observable both in 1896 and 1904. Columns (3) and (4) only include firms used in the lower bound estimation sample. Standard errors are clustered by product. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7 Competition and Market thickness

Unlike interventions that provide individual firms with targeted access to new buyers or suppliers, trade fairs bring together large numbers of firms simultaneously. There are two potential forces at play: firms may find potential buyers but also compete for attention. To examine this, we estimate heterogeneous treatment effects based on the number of competitors firms faced at the trade fair as well as market thickness – the number of potential buyers.

To do this, for all firms in Paris, we leveraged the product subcategory⁷ definition determined for all exhibitors. Some of the largest categories were "Wines and wine spirits" or "Mining, mining operations, and quarries", while examples for smaller ones include "Greenhouse Plants" or "Spinning and rope-making equipment and processes". The Paris catalog listed firms by these categories. There are 72 subcategories that included Hungarian firms and are relevant for private enterprises.

We construct two main measures of competition and buyers.

First, regarding competition, we measure the number of Hungarian competitors as the total number of Hungarian firms in the same Paris subcategory. Hungarian firms have known each other, and have little to gain in terms of business in Paris. A larger number of Hungarian exhibitors can intensify competition (more rivals in the same niche), but it could also have a positive effect if a stronger Hungarian presence helps attract more foreign buyers to that subcategory.

Second, we construct a buyers measure based on non-Hungarian exhibitors. Here we focus on

⁷Within each major industries of the time like Mines and metallurgy, there were several subcategories denoting finer product groups.

the product categories where potential buyers of the firm’s product are likely to be found. Using AI-assisted classification, we identify product categories that plausibly contain buyers for a given firm’s product, explicitly excluding the firm’s own product category. We then sum the number of exhibitors across all these “buyer” categories. This gives a measure of how many potential non-Hungarian buyers a firm is exposed to at the exhibition.

There is a large variation in both the number of Hungarian competitors (mean is 44 with a range of 3 to 157) and international buyers (mean is 3,149 with a range of 92 to 13,874). When ranking subcategories, we can see that there are few Hungarian firms in “Electrochemistry” or “Silks and silk fabrics”, while domestic competition is intense in “Wines and wine spirits.” or “Equipment and processes of food industries.” In terms of international buyers, tough product categories are “Hunting weapons” or “Musical instruments”, while we can expect many buyers in “Flour products and their derivatives.”, or “Mining, mining operations, and quarries.”

We estimated the following regression, which is an extension of the regression described in Equation 4 with the interaction terms for competition measures and the calculations for potential buyers:

$$\Delta y_i = \beta \text{exhibitor}_i + \gamma \text{exhibitor}_i \times \text{competition}_k + \delta \text{exhibitor}_i \times \text{buyer}_k + \text{BoundGroupFE}_j + FE_p + y_{i,pre} + \epsilon_i \quad (5)$$

where, outcomes and fixed effects are defined as for Equation 4. $\gamma \text{exhibitor}_i \times \text{competition}_k$ represents the effect of competition in Paris subcategory k on the outcomes for firm i if i is a Paris exhibitor. $\delta \text{exhibitor}_i \times \text{buyer}_k$ measures the effect of the number of potential buyers for Paris subcategory k for firm i if i is a Paris exhibitor.

Results are presented in Table 9 using the lower bound estimation method for export capacity. Competition and buyer market size are both measured as ln number of firms, and interacted with Paris indicator. The difference between the two columns is having all buyers in column (1) vs filtering out Hungarian buyers (in other product categories) in others.

Table 9: Effect of competition and potential buyers (Change in export capability), 1898-1904

	(1)	(2)
Paris exhibitor * ln(Hungarian competition)	-0.0418* (0.0249)	-0.0428* (0.0250)
Paris exhibitor * ln(number of buyers)	0.0414** (0.0167)	
Paris exhibitor * ln(number of international buyers)		0.0434** (0.0175)
Product effects	Yes	Yes
Bound group dummies	Lower	Lower
Sample	Lower	Lower
Observations	2,721	2,721
R-squared	0.487	0.487

Notes: Only include firms used in the lower bound estimation sample. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

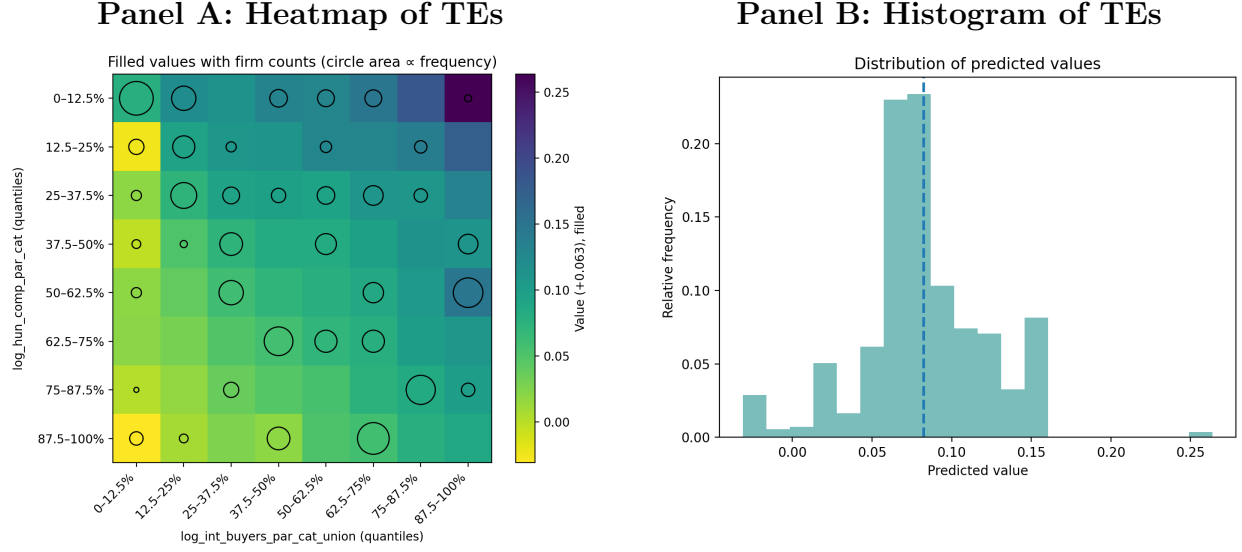
As Table 9 shows both mechanisms are at play: Hungarian firms that are part of product subcategories with many other Hungarian firms will benefit less of being in Paris. Attention is more divided, it is harder to make an impression. Hungarian firms in product subcategories that have many potential buyers benefit more.

Comparing firms at the bottom and top quartiles (the ln number of competitors at p25 is 2.77, and p75 is 4.11), the 1.34 log unit difference corresponds to a 5.74% (4.28×1.34) higher export probability or two-thirds (66%) of the average treatment effect of 8.63% (Table 5). Repeating the same comparison for the ln number of international buyers (p25 is 7.34, and p75 is 8.21), the 0.87 log unit difference corresponds to a 3.7% (4.3×0.87) higher export probability or almost half (43%) of the average treatment effect. Both these variations are large, and for the upper bound estimates, they would be even larger.

Figure 5 shows the heterogeneity of the treatment effect. As both panels confirm there is a large variation, the treatment varies between -0.02 and 0.25. There is only a very small segment of firms where the Paris exhibition has basically no effect (or even slightly negative point estimates), and a very small case for extremely high effect (where few Hungarian competitor and many potential buyers coincide). Still, there is large variation between 5% and 15%, around the top right part of the joint distribution.

These results suggest that being in Paris has a heterogeneous treatment effect across product subcategories: firms with fewer domestic peers and more potential buyers are the ones able to benefit more in terms of export opportunities. We also learned that a small segment of firms

Figure 5: Heterogeneous treatment effects by competition and international buyers



Note. Using the distribution of firms by competition and international buyer (ln) values, we create an 8×8 grid based on a parametric model mapped onto the realized joint distribution, and estimate the heterogeneous treatment effect distribution by competition and international buyers. The left panel shows a heatmap of treatment effects; missing cells (with zero firms) are filled by iterative averaging of neighboring cells to obtain a smooth surface. Black hollow circles indicate the number of firms in each cell, with circle area proportional to cell frequency. The right panel shows a histogram of treatment-effect values, with the weighted mean indicated by a dashed line.

did not benefit at all. This result is important as it indicates search and matching frictions are important and competition and large buyer base could alter the cost-benefit analysis of being at a large event.

These patterns of heterogeneity provide insight into the specific nature of search and matching frictions that trade fairs address. Trade fairs reduce discovery and inspection costs by concentrating buyers and sellers in a single location for a limited time, effectively lowering the search radius. However, our results reveal two important dimensions of how this mechanism operates in practice. First, the negative effect of domestic competition demonstrates a congestion externality: when multiple similar sellers compete for buyers' limited attention and time, the individual benefit of reduced search costs diminishes. This suggests that search frictions are not eliminated. Second, the positive effect of potential buyer concentration indicates that matching frictions remain even after search costs are reduced. A thicker buyer market increases both the probability of finding a suitable match and the expected quality of that match (Lenoir et al., 2023).

8 Conclusion

This paper provides causal evidence that trade fairs can effectively promote exports by reducing search and matching frictions. Using the 1900 Paris Exhibition as a natural experiment, we show

that participation increased Hungarian firms' export probability by 4-10 percentage points and generated 15-23% employment growth over the following decade. Export probabilities were highest for firms in thick markets (many potential buyers) but with few domestic competitors.

Our methodological contribution—using predetermined quality measures and selection constraints to bound treatment effects—addresses a fundamental challenge in evaluating targeted policies. When perfect randomization is infeasible, our approach shows how researchers can leverage institutional details to obtain credible causal estimates. In comparison to existing causal bands, our method offer a substantially narrower range.

The mechanisms underlying these effects highlight important considerations for modern policy. We find that congestion effects from too many similar firms can offset the benefits of participation. This suggests that simply maximizing participation may be counterproductive; instead, policies should consider the competitive dynamics within product categories. At the same time, policy shall focus on sectors and products that can attract a large foreign buyer base.

Our findings have important implications for contemporary policy. Despite substantial public investment in these programs, rigorous causal evidence on their effectiveness remains limited. Our results show that trade fair participation can generate substantial and persistent benefits—increasing export probability and employment. However, the heterogeneity we document reveals important design considerations: congestion effects from too many similar firms can offset participation benefits. This suggests that export promotion policy should consider competitive dynamics within product categories rather than simply maximizing participation. For middle-income countries seeking to expand into global markets, facilitating concentrated face-to-face interactions between domestic producers and foreign buyers can generate substantial returns, but the composition of participants matters critically for effectiveness. Our findings tilts the balance to well organized large events that provide many opportunities for relative few supported companies.

References

- Alfaro, Laura, Cathy Ge Bao, Maggie X. Chen, Junjie Hong, and Claudia Steinwender**, “Omnia Juncta in Uno: Foreign Powers and Trademark Protection in Shanghai’s Concession Era,” NBER Working Paper 29721 2022.
- Allen, Treb**, “Information frictions in trade,” *Econometrica*, 2014, 82 (6), 2041–2083.
- Arkolakis, Costas, Federico Huneus, and Yuhei Miyauchi**, “Production Network Formation, Trade, and Welfare,” 2025.
- Assunção, Juliano, Robert McMillan, Joshua Murphy, and Eduardo Souza-Rodrigues**, “Optimal environmental targeting in the amazon rainforest,” *The Review of Economic Studies*, 2023, 90 (4), 1608–1641.
- Athey, Susan and Guido W Imbens**, “Identification and inference in nonlinear difference-in-differences models,” *Econometrica*, 2006, 74 (2), 431–497.
- Atkin, David, Amit K. Khandelwal, and Adam Osman**, “Exporting and Firm Performance: Evidence from a Randomized Experiment,” *The Quarterly Journal of Economics*, 2017, 132 (2), 551–615.
- Bergquist, Lauren F, Craig McIntosh, and Meredith Startz**, “Search costs, intermediation, and trade: Experimental evidence from Ugandan agricultural markets,” 2024.
- Bernard, Andrew B., Andreas Moxnes, and Karen Helene Ulltveit-Moe**, “Two-Sided Heterogeneity and Trade,” *The Review of Economics and Statistics*, 07 2018, 100 (3), 424–439.
- Bernard, Andrew B, Andreas Moxnes, and Yukiko U Saito**, “Production networks, geography, and firm performance,” *Journal of Political Economy*, 2019, 127 (2), 639–688.
- Biesebroeck, Johannes Van, Emily Yu, and Shenjie Chen**, “The impact of trade promotion services on Canadian exporter performance,” *Canadian Journal of Economics/Revue canadienne d’économique*, 2015, 48 (4), 1481–1512.
- Blundell, Richard, Amanda Gosling, Hidehiko Ichimura, and Costas Meghir**, “Changes in the Distribution of Male and Female Wages Accounting for Employment Composition Using Bounds,” *Econometrica*, 2007, 75 (2), 323–363.
- Broocks, Annette and Johannes Van Biesebroeck**, “The impact of export promotion on export market entry,” *Journal of International Economics*, 2017, 107, 19–33.
- Buus, Magnus T, Jakob R Munch, Joel Rodrigue, and Georg Schaur**, “Do Export Support Programs Affect Prices, Quality, Markups, and Marginal Costs? Evidence From a Natural Policy Experiment,” *Review of Economics and Statistics*, 2025, 107 (1), 172–187.
- Békés, Gábor, Mátyás Molnár, and Claudia Steinwender**, “Globalization and industrial policy in the Austro-Hungarian Monarchy,” 2026. mimeo, CEU.
- Cai, Jing, Wei Lin, and Adam Szeidl**, “Firm-to-Firm Referrals,” 2024.
- Carballo, Jerónimo and Christian Volpe Martincus**, “Is export promotion effective in developing countries? Firm-level evidence on the intensive and the extensive margins of exports,” *Journal of International Economics*, 2008, 76 (1), 89–106.
- Demir, Banu, Ana Cecilia Fieler, Daniel Yi Xu, and Kelly Kaili Yang**, “O-ring production networks,” *Journal of Political Economy*, 2024, 132 (1), 200–247.
- Dhyne, Emmanuel, Ayumu Ken Kikkawa, Xianglong Kong, Magne Mogstad, and Felix Tintelnot**, “Endogenous production networks with fixed costs,” *Journal of International Economics*, 2023, 145, 103841.

- Eaton, Jonathan, David Jenkins, James R Tybout, and Daniel Xu**, “Two-sided search in international markets,” Technical Report, National Bureau of Economic Research 2022.
- , **Samuel S Kortum, and Francis Kramarz**, “Firm-to-Firm Trade: Imports, Exports, and the Labor Market,” Working Paper 29685, National Bureau of Economic Research January 2022.
- Gerard, François, Miikka Rokkanen, and Christoph Rothe**, “Bounds on treatment effects in regression discontinuity designs with a manipulated running variable,” *Quantitative Economics*, 2020, 11 (3), 839–870.
- Görg, Holger, Michael Henry, and Eric Strobl**, “Grant support and exporting activity,” *The review of economics and statistics*, 2008, 90 (1), 168–174.
- Hanlon, W Walker**, “The persistent effect of temporary input cost advantages in shipbuilding, 1850 to 1911,” *Journal of the European Economic Association*, 2020, 18 (6), 3173–3209.
- Hiller, Sanne**, “Does export promotion work in Denmark? Evidence from a matching approach,” 2012.
- Ho, Kate and Adam M Rosen**, “Partial identification in applied research: benefits and challenges,” *Advances in Economics and Econometrics*, 2017, 2, 307–359.
- Jensen, Robert**, “The digital provide: Information (technology), market performance, and welfare in the South Indian fisheries sector,” *The Quarterly Journal of Economics*, 2007, 122 (3), 879–924.
- and **Nolan H Miller**, “Market integration, demand, and the growth of firms: Evidence from a natural experiment in India,” *American Economic Review*, 2018, 108 (12), 3583–3625.
- Juhász, Réka and Claudia Steinwender**, “Spinning the web: The impact of ICT on trade in intermediates and technology diffusion,” Technical Report, National Bureau of Economic Research 2018.
- and —, “Industrial Policy and the Great Divergence,” *Annual Review of Economics*, 2024, 16.
- Juhász, Réka**, “Temporary Protection and Technology Adoption: Evidence from the Napoleonic Blockade,” *American Economic Review*, 108 (11): 3339-76., 2018.
- Klein, Alexander, Max-Stephan Schulze, and Tamás Vonyó**, “How Peripheral was the Periphery? Industrialization in East Central Europe since 1870,” in “The Spread of Modern Industry to the Periphery since 1871,” Oxford University Press, 02 2017.
- Krolkowski, Pawel M and Andrew H McCallum**, “Goods-market frictions and international trade,” *Journal of International Economics*, 2021, 129, 103411.
- Kun, Gergő**, “Industrial Development Policies in the Austro-Hungarian Empire, Focusing on Hungary’s Industry from the Austro-Hungarian Compromise of 1867 to the World War I,” *SEA–Practical Application of Science*, 2019, 7 (19), 55–64.
- Lederman, Daniel, Marcelo Olarreaga, and Lucy Payton**, “Export Promotion Agencies: Do They Work?,” *Journal of Development Economics*, 2010, 91 (2), 257–265.
- Lee, David S**, “Training, wages, and sample selection: Estimating sharp bounds on treatment effects,” *The Review of Economic Studies*, 2009, 76 (3), 1071–1102.
- Lenoir, Clémence, Julien Martin, and Isabelle Mejean**, “Search frictions in international goods markets,” *Journal of the European Economic Association*, 2023, 21 (1), 326–366.
- Lim, Kevin**, “Firm-to-firm Trade in Sticky Production Networks,” 2017 Meeting Papers 280, Society for Economic Dynamics 2017.
- Manski, Charles F**, *Partial identification of probability distributions*, Springer, 2003.

- Manski, Charles F. and John V. Pepper**, “Monotone Instrumental Variables: With an Application to the Returns to Schooling,” *Econometrica*, 2000, 68 (4), 997–1010.
- Martincus, Christian Volpe and Jerónimo Carballo**, “Beyond the average effects: The distributional impacts of export promotion programs in developing countries,” *Journal of Development Economics*, 2010, 92 (2), 201–214.
- McKinsey & Company**, “With Artificial Intelligence, Find New Suppliers in Days, Not Months,” 2021. Authors: Mark Baptista, Mike Gordon, Christine Herrmann, and Andrew Pratt. Accessed: 2025-08-01.
- Miyauchi, Yuhei**, “Matching and Agglomeration: Theory and Evidence from Japanese Firm-to-Firm Trade,” *Econometrica*, 2024, 92 (6), 1869–1905.
- Morales, Eduardo, Gloria Sheu, and Andrés Zahler**, “Extended Gravity,” *The Review of Economic Studies*, 02 2019, 86 (6), 2668–2712.
- Moser, Petra**, “How Do Patent Laws Influence Innovation? Evidence from Nineteenth-Century World’s Fairs,” *American Economic Review*, September 2005, 95 (4), 1214–1236.
- , “Innovation without Patents: Evidence from World’s Fairs,” *The journal of law economics*, 2012, 55 (1), 43–74.
- Munch, Jakob and Georg Schaur**, “The effect of export promotion on firm-level performance,” *American Economic Journal: Economic Policy*, 2018, 10 (1), 357–387.
- Munch, J.R., M. Buus, J. Rodrigue, and G. Schaur**, “Do Export Support Programs affect Prices, Quality, Markups and Marginal Costs? Evidence from a Natural Policy Experiment,” *Review of Economics and Statistics*, forthcoming.
- Rambachan, Ashesh and Jonathan Roth**, “A more credible approach to parallel trends,” *Review of Economic Studies*, 2023, 90 (5), 2555–2591.
- Romero, Dario A**, “An Empire Lost: Spanish Industry and the Effect of Colonial Markets on Trade and Innovation,” 2023.
- Schminke, Annette and Johannes Van Biesebroeck**, “Using export market performance to evaluate regional preferential policies in China,” *Review of World Economics*, 2013, 149 (2), 343–367.
- Steinwender, Claudia**, “Real Effects of Information Frictions: ”When the States and the Kingdom became United”,” *American Economic Review*, 2018, 108 (3), 657–696.
- Székely, Miklós**, “The Resetting of the Main Historical Group from the Millennium Exhibition to the Paris Universal Exhibition of 1900,” *Ephemeral Architecture in Central Eastern Europe in the 19th and 20th centuries*, 2015, pp. 33–50.
- Tamer, Elie**, “Partial identification in econometrics,” *Annual Review of Economics*, 2010, 2 (1), 167–195.
- Tamília, Robert**, “World’s Fairs and the Department Store 1800s to 1930s,” in “Proceedings of the Conference on Historical Analysis and Research in Marketing,” Vol. 13 2007, pp. 228–240.

A Manski Bounds

This appendix reviews Manski’s partial identification approach and demonstrates how our award-based bounds improve upon worst-case bounds.

A.1 Manski bounds: idea and setup

Our goal is to learn about the average treatment effect (ATE)

$$\text{ATE} = E[Y(1) - Y(0)]$$

when we only observe

$$Y = DY(1) + (1 - D)Y(0), \quad D \in \{0, 1\},$$

and we do not assume random assignment, selection-on-observables, instrumental variables, or other strong identifying assumptions.

Manski’s insight is to work under minimal assumptions, requiring only that Y is bounded, such as $Y \in [0, 1]$. We cannot point-identify the ATE, but we can find **sharp bounds**:

$$\underline{\text{ATE}} \leq \text{ATE} \leq \overline{\text{ATE}}.$$

These *Manski bounds* (or *worst-case bounds*) provide the tightest possible interval consistent with the data and boundedness assumption.

A.1.1 Worst-case Manski bounds

Consider a binary treatment $D \in \{0, 1\}$ and outcome $Y \in [0, 1]$ (e.g., probability or indicator). Potential outcomes $Y(1)$ and $Y(0)$ are never jointly observed.

The observed quantities are:

$$\mu_1 = E[Y \mid D = 1] = E[Y(1) \mid D = 1], \quad \mu_0 = E[Y \mid D = 0] = E[Y(0) \mid D = 0],$$

and treatment probabilities:

$$p_1 = P(D = 1), \quad p_0 = 1 - p_1.$$

The unknown counterfactual means

$$E[Y(1) \mid D = 0] \quad \text{and} \quad E[Y(0) \mid D = 1]$$

are never observed and, under only $Y \in [0, 1]$, can lie anywhere in $[0, 1]$.

A.1.2 Manski’s worst-case ATE bounds

The ATE can be decomposed as:

$$\text{ATE} = E[Y(1)] - E[Y(0)] = p_1 E[Y(1) \mid D = 1] + p_0 E[Y(1) \mid D = 0] - p_1 E[Y(0) \mid D = 1] - p_0 E[Y(0) \mid D = 0].$$

Define the unknown counterfactual means as

$$\theta_1 = E[Y(1) \mid D = 0] \quad \text{and} \quad \theta_0 = E[Y(0) \mid D = 1].$$

Substituting observed quantities yields:

$$\text{ATE} = p_1\mu_1 + p_0\theta_1 - p_1\theta_0 - p_0\mu_0.$$

Under $Y \in [0, 1]$, we have $0 \leq \theta_1 \leq 1$ and $0 \leq \theta_0 \leq 1$. Choosing extreme values to minimize and maximize the ATE gives:

$$\begin{aligned}\underline{\text{ATE}} &= p_1(\mu_1 - 1) - p_0\mu_0, \\ \overline{\text{ATE}} &= p_1\mu_1 + p_0(1 - \mu_0).\end{aligned}$$

Any ATE outside $[\underline{\text{ATE}}, \overline{\text{ATE}}]$ is incompatible with the data and assumption $Y \in [0, 1]$. These bounds are sharp.

A.1.3 Tightening the bounds

The Manski bounds are very conservative, making almost no structural assumptions. The interval

$$\underline{\text{ATE}} \leq \text{ATE} \leq \overline{\text{ATE}}$$

can be wide. When wide, it indicates the data alone cannot say much about the ATE without additional assumptions. Stronger claims require additional structure (e.g., unconfoundedness, instrumental variables).

One can tighten bounds with relatively weak assumptions. First, *Monotone Treatment Response* (MTR) posits $Y(1) \geq Y(0)$ for all units, plausible when treatment cannot harm. Second, *Monotone Treatment Selection* (MTS) posits treated units have weakly higher (or lower) potential outcomes.

Note that an additional idea could be instrumental variables to be incorporated within a partial identification framework.

Each assumption shrinks $[\underline{\text{ATE}}, \overline{\text{ATE}}]$, trading robustness for informativeness.

A.2 Estimated bounds

A.2.1 Bounds on ATE for Exporter Status in 1904

Figure A.1 shows the progression with MTR and MTS. From Manski bounds of $[-0.21, 0.79]$, MTR yields $[0.00, 0.79]$ (length ≈ 0.79), MTS yields $[-0.21, 0.39]$ (length ≈ 0.60), and their intersection gives $[0.00, 0.39]$ (length ≈ 0.39). Our award-based bounds of $[0.085, 0.14]$ (length ≈ 0.055) provide much sharper inference.

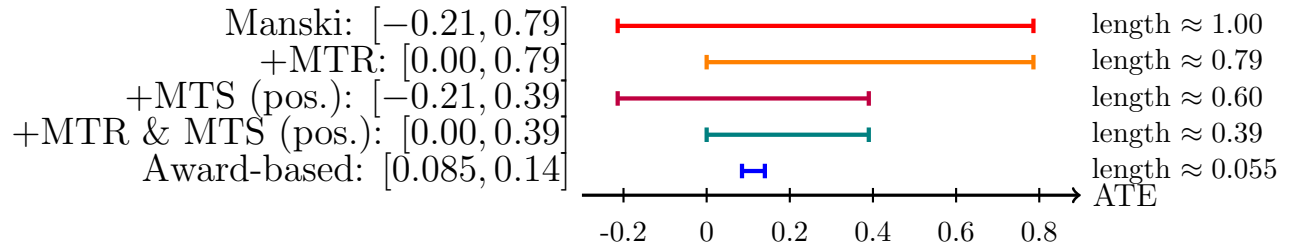


Figure A.1: Bounds on ATE for Exporter Status in 1904 with MTR and MTS

A.2.2 Bounds on ATE for Exporter Status in 1905

Figure A.2 shows how bounds change with MTR and MTS assumptions. Starting from Manski bounds of $[-0.16, 0.84]$, adding MTR yields $[0.00, 0.84]$ (length ≈ 0.84). Adding MTS with positive selection yields $[-0.16, 0.24]$ (length ≈ 0.40). Their intersection gives $[0.00, 0.24]$ (length ≈ 0.24), still leaving substantial uncertainty. Our award-based bounds deliver a much tighter interval around 0.07.

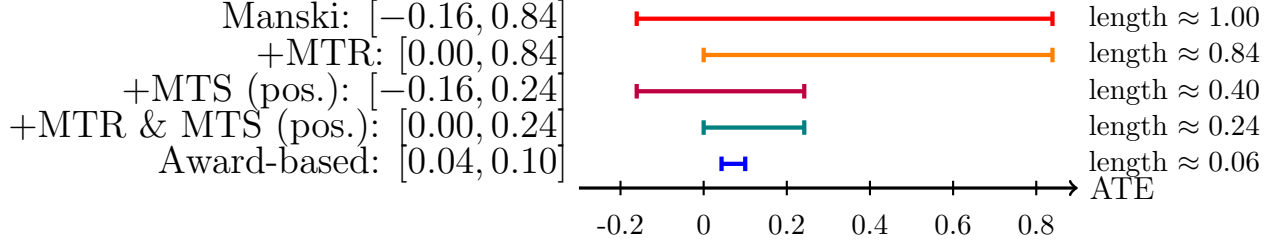


Figure A.2: Bounds on ATE for Exporter Status in 1905 with MTR and MTS

A.2.3 Summary of bounds on ATE

Tables A.1 and A.2 summarize ATE bounds under different identification strategies.

Manski worst-case bounds are extremely wide (length ≈ 1.00), spanning from substantially negative to very large positive values. Adding MTR or MTS individually provides some tightening, but intervals remain wide. MTR eliminates negative values; MTS with positive selection narrows the upper bound. Their intersection produces tighter bounds (length 0.24–0.39) but leaves considerable uncertainty.

Our award-based bounds dramatically outperform alternatives. For 1905, bounds of $[0.04, 0.10]$ (length 0.06) represent approximately 75% reduction versus MTR and MTS combined. For 1904, bounds of $[0.085, 0.14]$ (length 0.06) similarly dominate. Our bounds are strictly positive, narrow, and provide strong, precise evidence of treatment effects. This improvement comes from exploiting Budapest exhibition awards as ex-ante measures of export potential.

Table A.1: Bounds on ATE for Δ international exporter 1905

Outcome: Δ international exporter	ATE interval
Manski worst-case (no assumptions)	$[-0.16, 0.84]$
+ MTR ($Y(1) \geq Y(0)$)	$[0.00, 0.84]$
+ MTR & MTS ⁺ (pos. selection)	$[0.00, 0.028]$
Award-based bounds	$[0.04, 0.10]$

Table A.2: Bounds on ATE for Δ domestic trader or exporter 1904

Outcome: Δ domestic trader or exporter 1904	ATE interval
Manski worst-case (no assumptions)	$[-0.21, 0.79]$
+ MTR ($Y(1) \geq Y(0)$)	$[0.00, 0.79]$
+ MTR & MTS ⁺ (pos. selection)	$[0.00, 0.026]$
Award-based bounds	$[0.085, 0.14]$

B Data Appendix

This appendix documents the data collection, processing, and variable construction procedures for the analysis of Hungarian firms’ exhibition participation.

In this appendix, we discuss the key steps awhile acknowledging inherent limitations of historical administrative sources.

B.1 The 1896 Budapest Exhibition

Budapest Exhibition 1896 catalog Complete official catalog of the 1896 Millennial National Exhibition containing 4,626 exhibitor records across contemporary industrial categories (agriculture excluded). Source: *Az 1896-iki Ezredéves Országos Kiállítás általános katalógusa*, edited by Mudrony Soma, Ráth Károly, and Micseh Endre (Budapest: Kosmos Ny., 1896), shelf number 282.068. The catalog comprises 21 category-specific volumes, each with independent numbering systems.

Jury Awards 1896. Official jury evaluations from the Budapest exhibition, documented in *Az 1896-iki ezredéves országos kiállítás közleményei*, issue 32, pages 1229-1236 (shelf number H 2.008 FM3/4736). Awards categorized hierarchically: Grand Certificate, Grand Medal, Exhibition Medal, Certificate of Acknowledgement, and Contributors’ Medal.

State Awards 1896. Government-appointed committee awards documented in the same publication, comprising 100 gold medals and 200 silver medals distributed independently of jury evaluations.

Table B.1: Exhibition Awards Variables

Variable	Definition and Construction
<code>jury_award_1896</code>	Categorical variable for jury award level: Grand Certificate (highest), Grand Medal, Exhibition Medal, Certificate of Acknowledgement. Contributors’ Medal excluded from firm-level analysis.
<code>state_award_1896</code>	Binary indicators for government-appointed committee awards: gold medal, silver medal. Distributed independently of jury awards.
<code>jury_award_explanation_1896</code>	Textual explanation of award rationale (e.g., ”good design,” ”export performance,” ”new industry establishment”). Enables analysis of award criteria and firm characteristics.

B.2 The 1900 Paris Exhibition and Trade Fair

Paris Exhibition 1900 catalog Hungarian-language catalog of Magyar exhibitors at the 1900 Paris International Exhibition, documenting 2,665 participants with detailed category and subcategory classifications. Source: *Az 1900. Párisi Magyar Kiállítás kalauza: Magyarország ismertetése* (Budapest: Riegler Ny., 1900), shelf number 262.540. This catalog serves as the primary treatment indicator for the empirical analysis.

B.3 Data Digitization and Text Processing

B.3.1 Digitization Protocol

All primary sources accessed in printed format underwent systematic digitization: (1) photographic capture using Hungarian National Library equipment, (2) optical character recognition via Google Document AI, and (3) systematic error correction through comparison with original printed books.

B.3.2 Text Processing Pipeline

Raw OCR output processed using Python with regular expression libraries implementing:

1. **Character encoding standardization:** Conversion of Hungarian diacritical marks and historical orthographic conventions to Unicode UTF-8 format
2. **Field extraction:** Parsing of semi-structured text entries into discrete variables using pattern recognition algorithms tailored to each source’s formatting conventions
3. **Manual verification:** Systematic correction of automated extraction errors, with particular attention to firm names, addresses, and numerical data

B.4 Exhibition Participation and Awards

B.4.1 Exhibition Variables

Table B.2: Exhibition Participation Variables

Variable	Definition and Construction
exhibition_category_YYYY	Industry classification within exhibition framework. Original categorization preserved with harmonized coding for cross-temporal analysis.
exhibition_subcategory_YYYY	Detailed product-level classification within broader industry categories. Enables fine-grained analysis of exhibition participation patterns.
exhibited_product_YYYY	Specific products displayed at exhibition, distinct from firm’s full product portfolio. Captures strategic decisions regarding exhibition representation.

B.5 Entity Resolution and Panel Construction

B.5.1 Matching Methodology

Panel dataset construction required systematic entity resolution across five independently collected datasets. The matching procedure implemented:

Pairwise Matching Algorithm:

1. **String similarity:** Jaro-Winkler distance calculation for firm names with threshold optimization balancing precision and recall

2. **Geographic consistency:** Exact and approximate location matching using standardized municipal codes
3. **Product similarity:** Jaccard similarity coefficients for product portfolio overlap
4. **Temporal consistency:** Validation of firm characteristic evolution (employment, technology) for plausibility

Correspondence Tables: Five pairwise matching tables documenting confirmed firm linkages:

- 1896 Budapest Catalog ↔ 1898 Export Registry
- 1896 Budapest Catalog ↔ 1900 Paris Catalog
- 1896 Budapest Catalog ↔ 1904 Export Registry
- 1898 Export Registry ↔ 1904 Export Registry

Master Identification Table: Comprehensive mapping consolidating direct and indirect matches to enable multi-period panel construction with both balanced and unbalanced specifications.

B.5.2 Match Quality and Coverage

Table B.3: Entity Resolution Results

Dataset Pair	Total Potential Matches	Confirmed Matches	Match Rate (%)
1896 Budapest ↔ 1898 Export	$4,626 \times 1,894$	1,059	55.9
1896 Budapest ↔ 1900 Paris	$4,626 \times 2,665$	665	24.9
1896 Budapest ↔ 1904 Export	$4,626 \times 4,381$	957	21.8
1898 Export ↔ 1904 Export	$1,894 \times 4,381$	1,114	58.8

B.6 Sample Characteristics and Descriptive Statistics

B.6.1 Dataset Coverage and Temporal Scope

Table B.4: Dataset Characteristics

Dataset	Observations	Time Period	Processing Status	Key Innovation
Budapest 1896	4,626	1896	Complete*	Universal exhibition census
Export 1898	1,894	1897-1898	Complete	Export-ready classification
Paris 1900	2,665	1900	Complete	International treatment
Export 1904	4,381	1904-1906	Complete	Comprehensive registry

*Excludes education and agriculture categories

B.7 Measurement Limitations and Econometric Considerations

B.7.1 Historical Source Constraints

Selection Mechanisms: The 1894 registry’s restriction to self-reporting exporters introduces potential selection bias, addressed through sensitivity analysis using 1898 export-ready classifications.

Measurement Error: Firm-reported employment and capital measures subject to strategic reporting incentives and standardization challenges across heterogeneous enterprises.

Survival Bias: Panel analysis necessarily conditions on firm survival through the observation period, potentially understating treatment effects for marginal firms.

B.7.2 Entity Resolution Limitations

Several issues arose during the process that may generate both types of errors. We looked at matches manually but it remains possible.

Matching Precision: Name variations, mergers, and reorganizations

Geographic Mobility: Firm relocation between observations may complicate identification. (A systematic address tracking suggests limited geographic mobility for manufacturing enterprises.)

B.7.3 Variable Construction Issues

Industry Classification Harmonization: Evolving industrial classification systems across datasets necessitate concordance tables with potential misclassification of diversified firms.

Temporal Aggregation: Registry publication dates may not correspond precisely to data collection periods, introducing timing measurement error requiring robustness checks with alternative temporal specifications.

C Exhibition Rules and Institutional Framework

The credibility of our empirical strategy relies on understanding the institutional rules governing both exhibitions. The Budapest exhibition provided an objective measure of export potential through its independent jury system, while the Paris exhibition’s selection constraints created the variation necessary for identification.

C.1 Budapest 1896 Millennial Exhibition Rules

The 1896 Budapest exhibition was designed as a comprehensive showcase of Hungarian industry and served as a “trial run” for identifying firms with export potential. The exhibition attracted over 9,000 exhibitors across all sectors, with mandatory evaluation of all commercial exhibits by an independent jury system.

C.1.1 Jury Independence and Structure

The jury operated independently from any future selection processes. According to the official regulations, jury members were selected through a mixed process: half elected by exhibitors within each industry group, half appointed by the exhibition committee in consultation with relevant ministries. The jury consisted of 480 members across 34 specialized industry groups, with each group electing its own president, vice-president, and rapporteur by secret ballot and simple majority.

Crucially, jury members and their associated firms were explicitly excluded from receiving awards in their respective groups, ensuring objective evaluation. Experts serving in advisory roles were similarly excluded from awards in the specific groups where they provided assistance. State authorities, institutions, and enterprises were also ineligible for awards, with their items marked with appropriate labels.

C.1.2 Award Categories and Hierarchy

The jury could award five distinct categories of recognition in descending order of prestige:

1. **Honorary Diploma:** The highest award, limited to a maximum of 100, given to exhibitors who had earned “national merit” through their contributions. These were awarded by the jury council based on group jury proposals.
2. **Millennial Grand Medal:** Awarded to exhibitors who “demonstrated exceptional advancement in their field, elevated their industry, introduced new types of production or methods, contributed to mass production or exports, and enhanced the prestige of the exhibition through the excellence of their exhibits.”
3. **Exhibition Medal:** Given to exhibitors in any group who demonstrated merit worthy of distinction.
4. **Certificate of Recognition:** Granted to exhibitors not eligible for higher awards but whose work deserved acknowledgment.
5. **Collaborators’ Medal:** For individuals who played significant roles in creation, design, engineering, decoration, or presentation of exhibited items.

Except for honorary diplomas, there was no limit to the number of medals and certificates awarded. Each medal was accompanied by a certificate listing the specific merits upon which the award was based.

C.1.3 State Awards

Independently of the jury awards, the Hungarian government established Millennial State Medals to honor achievements in public economy and culture: 100 gold medals and 200 silver medals. These were awarded by the Minister of Trade based on reasoned proposals from the jury council, providing a separate recognition track from the jury evaluation system.

C.1.4 Award Criteria and Export Focus

The regulations explicitly specified export capability as a primary evaluation criterion. Standard citation categories included: “good/excellent export capability,” “introducing a new industry in Hungary,” “good/excellent work, effort, or performance,” and “good/excellent progress or innovation.” The emphasis on export potential was central to the exhibition’s role as a trial run for identifying firms suitable for international representation.

C.1.5 Evaluation Process

All exhibited items underwent mandatory evaluation unless the exhibitor explicitly declined within a set deadline. The jury conducted systematic on-site inspections and decided for each exhibitor: (1) whether they should receive a distinction and why, (2) which award should be granted, and (3) what specific citation should be used.

Group jury decisions were made by majority vote, with the president’s vote deciding in case of ties. When decisions were not unanimous, minority opinions could be recorded upon request. The evaluation process was conducted under strict confidentiality, with no appeals permitted against jury council decisions.

C.2 Paris 1900 Exhibition Rules and Selection Process

The Paris exhibition operated under a sophisticated three-tier international jury system, but Hungarian participation was determined by a separate domestic selection process subject to multiple constraints beyond export potential.

C.2.1 Scale and Commercial Purpose

The 1900 Paris Exposition Universelle was explicitly designed as the largest international commercial gathering to date, featuring 76,000 exhibitors from 56 countries across 121 product sub-categories. The exhibition’s commercial objectives were formally recognized, with contemporary documentation describing it as providing “favorable opportunities for establishing commercial connections” and facilitating face-to-face meetings between international buyers and sellers.

C.2.2 Jury Structure and Award System

The Paris jury system operated through three hierarchical levels:

1. **Class Juries:** Foundation level, with approximately one titular member per 60 exhibitors, plus substitute members (limited to one-third of titular membership). Composed of both French jurors (appointed by ministerial decree) and foreign jurors (designated by national commissioners).

2. **Group Juries:** Intermediate level including presidents, vice-presidents, secretaries, plus all class jury officers, responsible for coordinating and harmonizing related categories.
3. **Superior Jury:** Highest authority including ministers, group jury leaders, and national delegates from countries with over 500 exhibitors, with final decision-making power over all awards.

C.2.3 Paris Award Categories

The Paris exhibition featured five award categories, all bearing signatures of the Minister of Commerce and Commissioner General:

1. **Grand Prix:** Highest distinction
2. **Gold Medal:** Second-tier recognition
3. **Silver Medal:** Third-tier recognition
4. **Bronze Medal:** Fourth-tier recognition
5. **Honorable Mention:** Fifth-tier recognition

Additionally, commemorative diplomas were available for contributors, exhibition staff, committee members, and jurors themselves.

C.2.4 Award Eligibility and Restrictions

Strict eligibility restrictions governed award consideration to ensure fairness. Jurors (both titular and supplementary) along with their associated firms were completely excluded from award consideration. Experts and associates providing advisory services to juries were excluded from awards in the specific classes where they assisted. Exhibitors could potentially receive awards in multiple classes but were limited to one award per class, with only the highest award applied if multiple juries evaluated the same object.

C.2.5 Evaluation Timeline and Process

The evaluation followed a structured timeline: class juries submitted their lists by June 30, group juries completed their review by July 31, and the superior jury finalized all decisions for an awards ceremony in late August or early September 1900. All jury deliberations were conducted under strict confidentiality requirements. Class rapporteurs were required to submit comprehensive reports within six months highlighting progress since the 1889 exhibition.

C.2.6 Hungarian Selection Constraints

While the primary goal was promoting exports by selecting high-potential firms, internal government records reveal three additional constraints that prevented perfect sorting based solely on export potential:

1. **Space Limitations:** Limited pavilion space in Paris restricted total participation numbers.

2. **Representational Balance:** The selection committee sought balanced representation across industries and regions to present a comprehensive picture of Hungarian production capabilities.
3. **Transport Costs:** High costs for shipping exhibits, particularly heavy machinery and bulky products, influenced selection decisions and created logistical constraints.

C.2.7 Product Category Organization

Exhibitors were classified into 18 broad categories and 121 subcategories based on their specific exhibits rather than their full product portfolios. This classification system determined competitive exposure at the exhibition, with subcategory sizes varying from hundreds to thousands of exhibitors. Categories ranged from narrowly defined products (e.g., “shoes”) to broader industrial classifications (e.g., “chemicals”).

C.2.8 Commercial Networking Framework

The exhibition operated from April to November 1900, providing extended opportunities for commercial relationship formation. The 48 million visitors included merchants, industrialists, and potential customers, creating an unprecedented concentration of international commercial contacts. Contemporary accounts emphasize active commercial negotiations occurring alongside product displays throughout the exhibition grounds.

These institutional details underscore that Budapest awards provide a credible measure of export potential determined by independent expert evaluation using explicit export-focused criteria, while Paris selection involved strategic balancing across multiple objectives, creating the selection patterns we exploit for identification.